

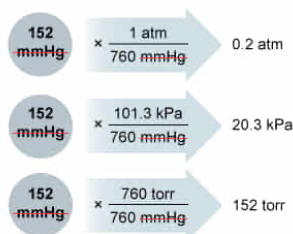
Which of the following pressure measurements is equivalent to a pressure of 152 mmHg?

- I. 0.2 atm
 - II. 20.3 kPa
 - III. 76 torr
- ☒ A. I and II only [73%]
☐ B. I and III only [12%]
☐ C. II and III only [9%]
☐ D. I, II, and III [5%]

Correct
72% answered correctly

Explanation:

Pressure measurement conversions



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Pressure P is defined as the amount of force F applied per unit of area A , which is expressed mathematically as

$$P = \frac{F}{A}$$

In the SI system, the standard unit of pressure is the **pascal (Pa)**, where 1 pascal is defined as the pressure exerted by a force of 1 newton (N) applied over an area of 1 square meter (m^2):

$$1 \text{ Pa} = 1 \frac{\text{N}}{\text{m}^2}$$

Because the pascal is a relatively small unit of pressure, measurements in pascals are often stated in kilopascals (kPa). The pascal unit is useful for relating pressure to other **derived units** that involve **base units** of mass, length, and time; however, measurements of pressure are often stated in other common units such as atmospheres (**atm**), millimeters of mercury (**mmHg**), or **torr**. These common units of pressure are related as follows:

$$1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ torr}$$

These relationships between units of pressure can be used to make **conversion factors** to convert pressure measurements from one unit to another. Accordingly, a pressure of 152 mmHg can be converted and expressed in units of atm, torr, or kilopascals, as follows:

$$\begin{aligned}
 152 \text{ mmHg} &\times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 0.2 \text{ atm} \\
 152 \text{ mmHg} &\times \frac{101.3 \text{ kPa}}{760 \text{ mmHg}} = 20.3 \text{ kPa} \\
 152 \text{ mmHg} &\times \frac{760 \text{ torr}}{760 \text{ mmHg}} = 152 \text{ torr}
 \end{aligned}$$

Therefore, a pressure of 152 mmHg is equivalent to 0.2 atm and 20.3 kPa (**Numbers I and II**).

(Number III) *Torr* is just another name for the unit *millimeters of mercury* (mmHg). As such, a pressure measurement given in mmHg will be the same numeric value in units of torr. A pressure of 76 torr is less than (not equivalent to) a pressure of 152 mmHg.

Educational objective:

Pressure measurements may be stated in a variety of units, including pascals (Pa), kilopascals (kPa), atmospheres (atm), millimeters of mercury (mmHg), and torr. These units can be converted from one to another by using appropriate conversion factors, given that $1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ torr} = 101,325 \text{ Pa} = 101.325 \text{ kPa}$.

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